# msvtx_trackSoftSumNTracks

MC vs data · overview · rank 64 / 98



rank 64 / 98 by endcap D_equal

used by: neither NN list

rank-identical columns merged into this one:
    msvtx_trackSoftSumPtStats/nTracks

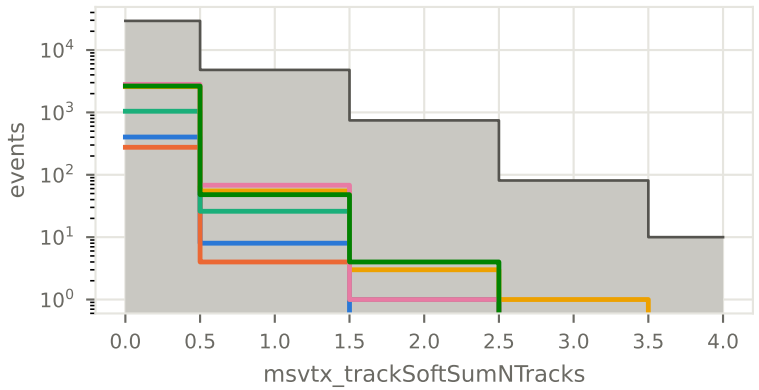
| region | D_equal | noise | D_pooled | 125/15 | 125/55 | 200/50 | 400/100 | 600/150 | 1000/275 | n_bkg | rank |
|---|---|---|---|---|---|---|---|---|---|---|---|
| endcap | 0.140 | 0.022 | 0.139 | 0.142 | 0.143 | 0.136 | 0.139 | 0.137 | 0.142 | 34,970 | 64/98 |
| barrel | 0.110 | 0.051 | 0.115 | 0.106 | 0.088 | 0.118 | 0.119 | 0.115 | 0.116 | 1,089 | 74/98 |
| all | 0.137 | 0.018 | 0.138 | 0.138 | 0.132 | 0.137 | 0.139 | 0.137 | 0.140 | 36,059 | 81/98 |

# msvtx_trackSoftSumNTracks

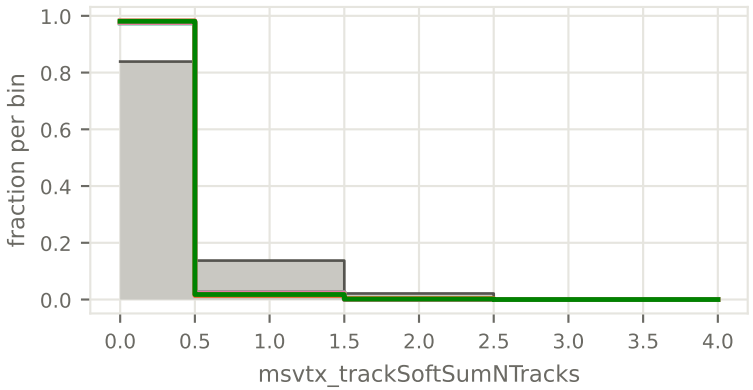
region: ENDCAP · rank 64 / 98

D_equal 0.140 (noise 0.022) D_pooled 0.139 min 0.136 (mH200_mS50) max 0.143 (mH125_mS55) bkg n = 34,970 sig n = 9,996

1a · each mass point vs background · raw

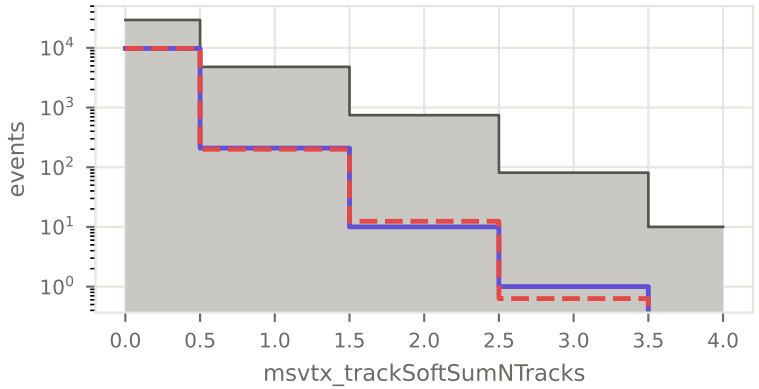



1b · each mass point vs background · normalized

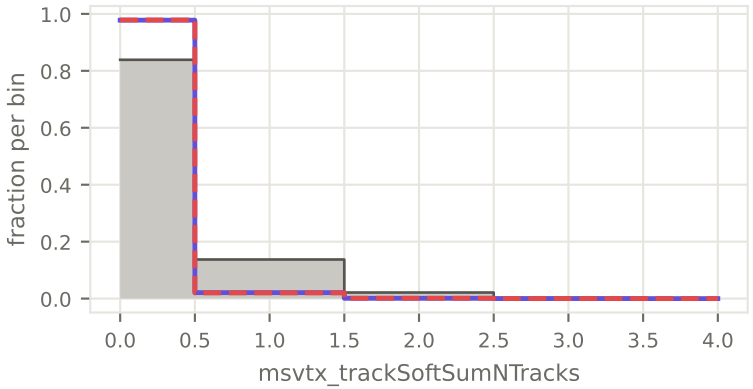



1 · each mass point
background n 34,970
mH125_mS15 D 0.142 n 411
mH125_mS55 D 0.143 n 281
mH200_mS50 D 0.136 n 1,071
mH400_mS100 D 0.139 n 2,648
mH600_mS150 D 0.137 n 2,882
mH1000_mS275 D 0.142 n 2,703

2a · combined signal vs background · raw

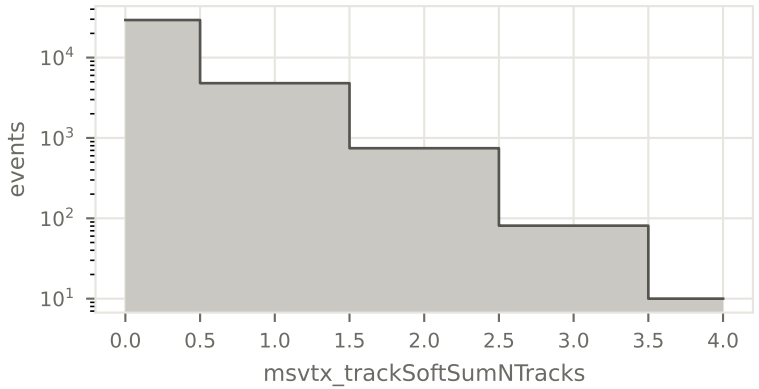



2b · combined signal vs background · normalized

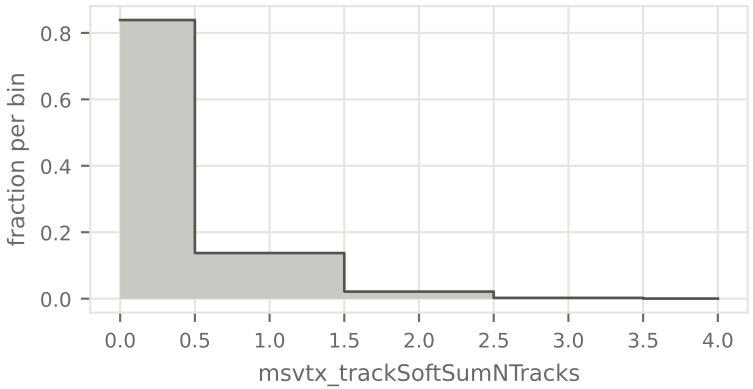



2 · combined signal
background n 34,970
combined, pooled D 0.139 n 9,996
equal weight per mass D 0.140

3a · background alone · raw




3b · background alone · normalized




3 · background alone
NaN 0.0% −999 0.0%
background n 34,970

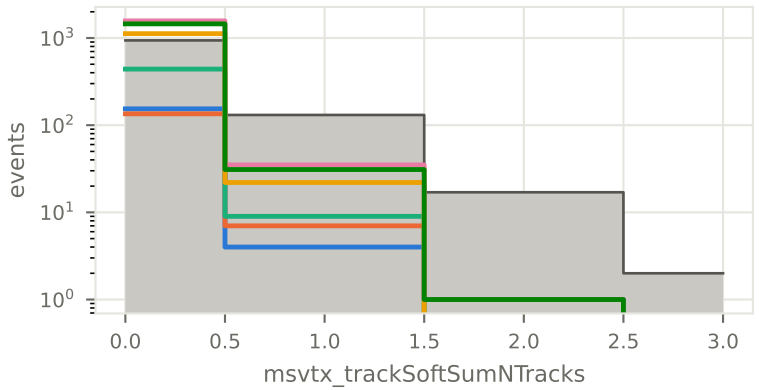
a = raw (log y) b = normalized · edge bins include under/overflow · NaN and -999 removed

# msvtx_trackSoftSumNTracks

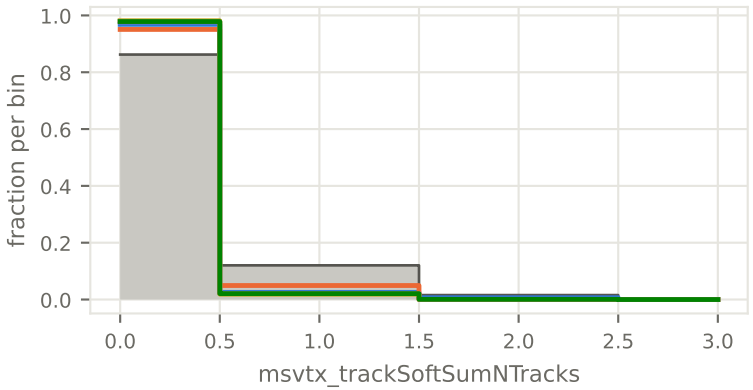
region: BARREL · rank 64 / 98

D_equal 0.110 (noise 0.051) D_pooled 0.115 min 0.088 (mH125_mS55) max 0.119 (mH400_mS100) bkg n = 1,089 sig n = 4,987

1a · each mass point vs background · raw

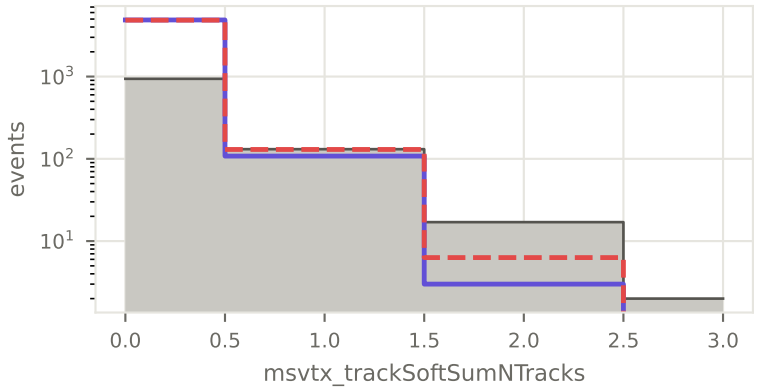



1b · each mass point vs background · normalized

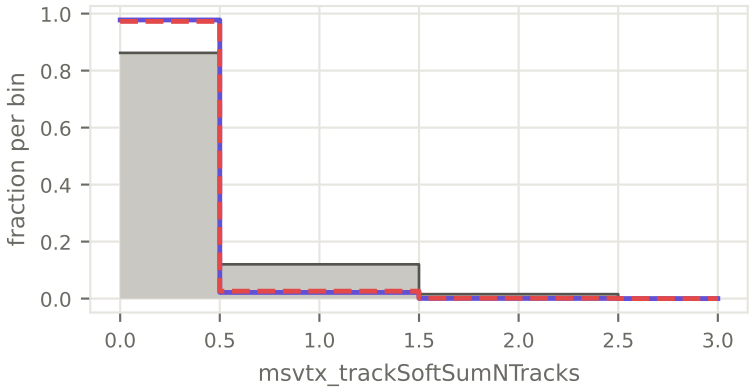



1 · each mass point
background n 1,089
mH125_mS15 D 0.106 n 159
mH125_mS55 D 0.088 n 142
mH200_mS50 D 0.118 n 449
mH400_mS100 D 0.119 n 1,144
mH600_mS150 D 0.115 n 1,608
mH1000_mS275 D 0.116 n 1,485

2a · combined signal vs background · raw

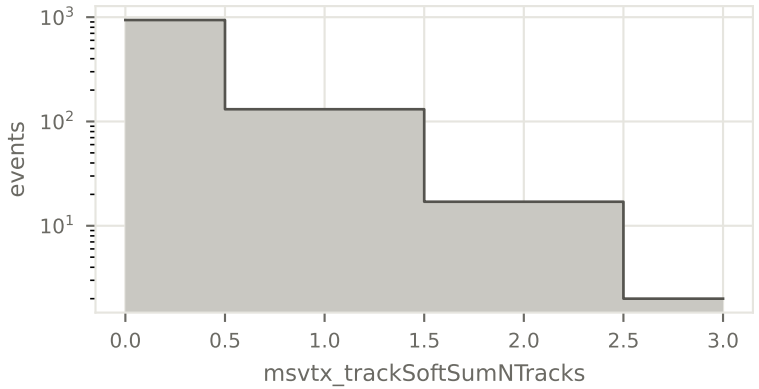



2b · combined signal vs background · normalized

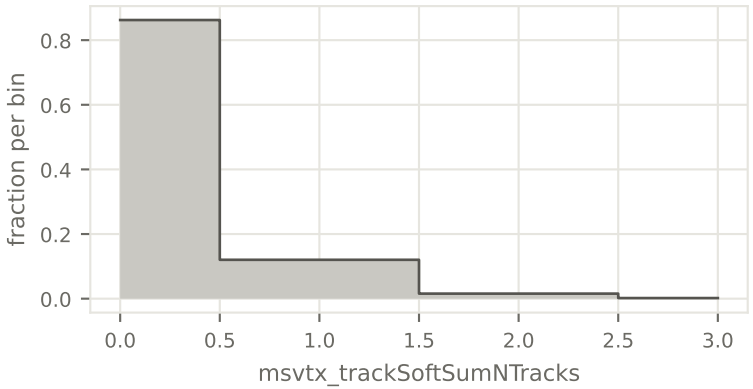



2 · combined signal
background n 1,089
combined, pooled D 0.115 n 4,987
equal weight per mass D 0.110

3a · background alone · raw




3b · background alone · normalized




3 · background alone
NaN 0.0% −999 0.0%
background n 1,089

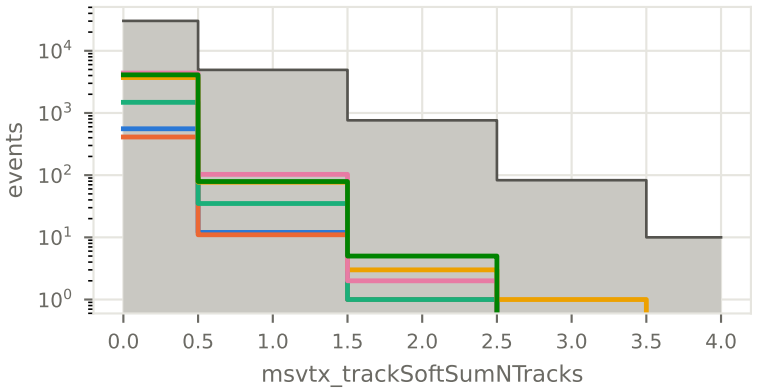
a = raw (log y) b = normalized · edge bins include under/overflow · NaN and -999 removed

# msvtx_trackSoftSumNTracks

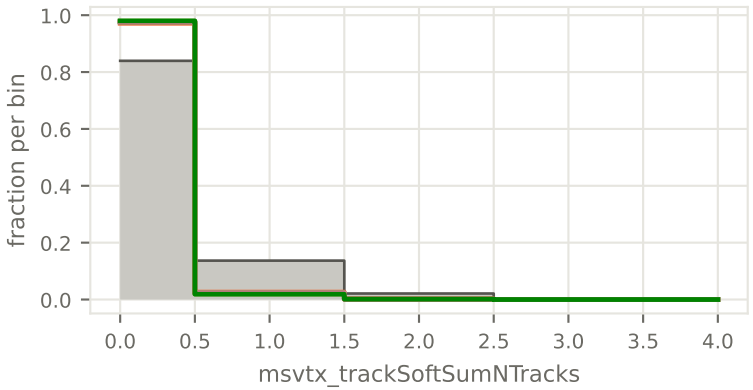
region: ALL · rank 64 / 98

D_equal 0.137 (noise 0.018)  D_pooled 0.138  min 0.132 (mH125_mS55)  max 0.140 (mH1000_mS275)  bkg n = 36,059  sig n = 14,983

1a · each mass point vs background · raw

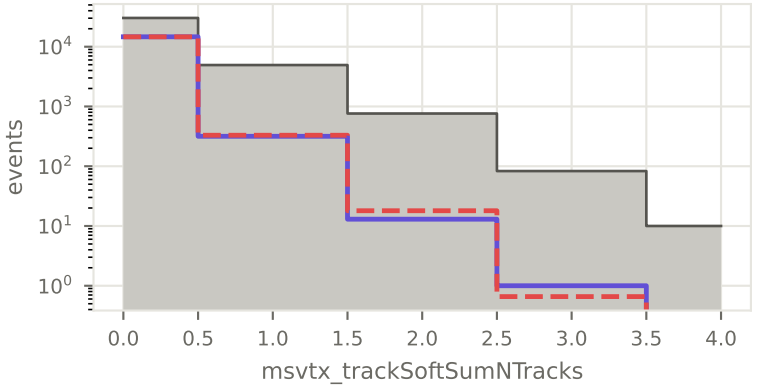



1b · each mass point vs background · normalized

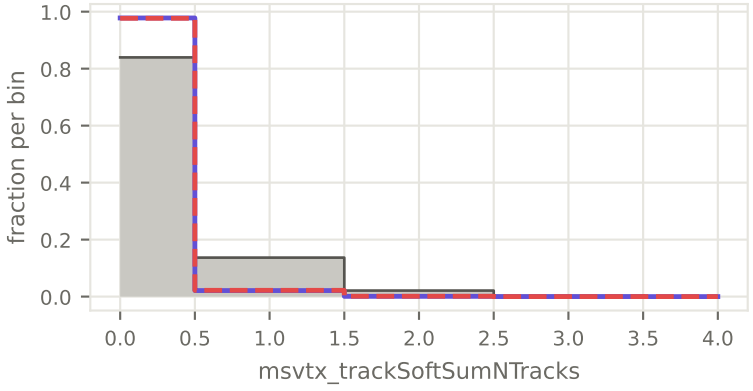



1 · each mass point
background n 36,059
mH125_mS15 D 0.138 n 570
mH125_mS55 D 0.132 n 423
mH200_mS50 D 0.137 n 1,520
mH400_mS100 D 0.139 n 3,792
mH600_mS150 D 0.137 n 4,490
mH1000_mS275 D 0.140 n 4,188

2a · combined signal vs background · raw

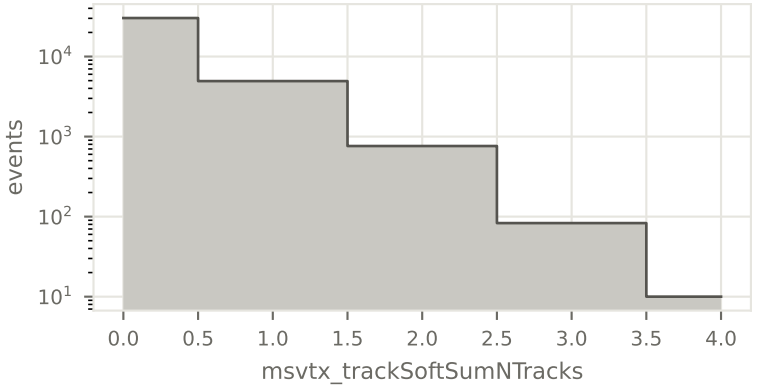



2b · combined signal vs background · normalized

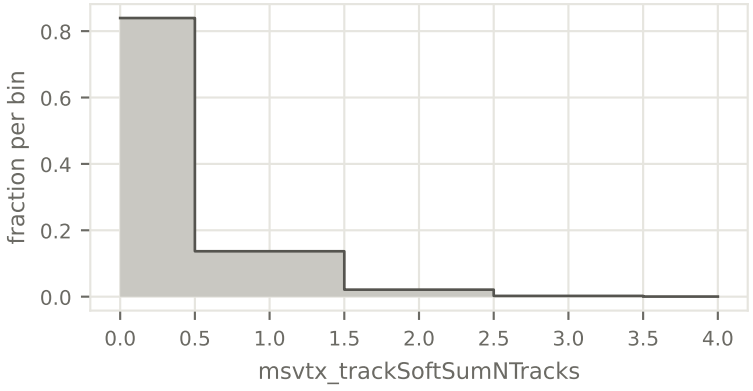



2 · combined signal
background n 36,059
combined, pooled D 0.138 n 14,983
equal weight per mass D 0.137

3a · background alone · raw




3b · background alone · normalized




3 · background alone
NaN 0.0%  −999 0.0%
background n 36,059

a = raw (log y)  b = normalized · edge bins include under/overflow · NaN and -999 removed